Cost of Living vs. Country Development

Data Collection and Preparation - Project Report

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Sources: Numbeo Cost of Living Index by Country (2026 Mid-Year) and World Bank API (2022)

1. Objective and approach

This project examines how cost of living is connected with GDP per capita, life expectancy and unemployment. We combined one static web source with one API source and used Python, requests, BeautifulSoup4, pandas, matplotlib and seaborn.

2. Data collection and preparation

Numbeo: We checked robots.txt, then used requests and BeautifulSoup to extract the country ranking table. We selected the required columns, renamed them, removed extra spaces, converted index values to numeric types, and checked missing values and duplicates. The cleaned dataset contains **152 countries and 6 columns**.

World Bank: We collected 2022 GDP per capita, life expectancy and unemployment. Aggregates were removed, numeric types were checked, and rows with missing indicator values were excluded. Country names were harmonized before an inner merge. The final dataset contains **127 countries and 10 columns**, with no missing values or duplicates.

3. Descriptive statistics and correlation

Variable	Mean	Median	Min	Max
GDP per capita (US$)	20,779	8,329	357	123,720
Life expectancy (years)	74.8	75.2	54.1	84.0
Unemployment (%)	6.6	5.0	0.13	33.3
Cost of living index	45.8	41.6	18.1	109.8
Rent index	16.4	13.5	2.4	68.3
Groceries index	47.5	43.3	18.8	116.3
Restaurant price index	41.9	38.6	10.9	108.1

GDP per capita is strongly right-skewed: its mean is much higher than its median. Pearson correlation shows a **strong positive relationship between GDP per capita and cost of living (r = 0.74)**. The strongest GDP relationship is with **rent (r = 0.83)**; restaurant prices (r = 0.79) and groceries (r = 0.72) are also strongly related. Life expectancy has a moderate positive relationship with cost of living (r = 0.56), while unemployment has only weak negative correlations with prices (about r = -0.17 to -0.27).

4. Visual results

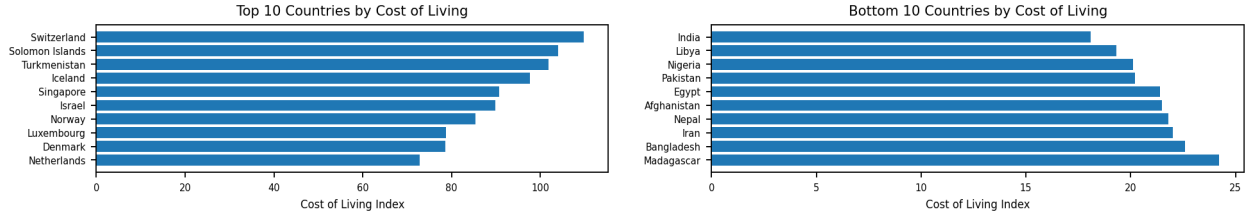


Figure 1. Top and bottom 10 countries by cost of living index.

Switzerland has the highest cost-of-living index (109.8), followed by Solomon Islands and Turkmenistan. India is the lowest (18.1), followed by Libya, Nigeria and Pakistan. Solomon Islands and Turkmenistan are notable outliers: their high prices do not follow the general development pattern and may reflect limited crowd-sourced coverage or country-specific factors.

5. Key findings, limitations and conclusion

Key finding: wealth and prices generally move together, and rent has the strongest relationship with GDP. Higher life expectancy is also associated with higher prices, while unemployment is only weakly related. **Limitations:** correlation does not prove causation; World Bank data is from 2022 while Numbeo is from 2026 Mid-Year; Numbeo is crowd-sourced; and only 127 of 152 Numbeo countries remain after cleaning and matching. **Conclusion:** the final analysis supports a strong positive relationship between country development and cost of living, but several outliers show that GDP alone does not explain prices.

Project files

Notebook: Cost_of_Living_vs_Country_Development_Final.ipynb
GitHub: https://github.com/kenzhebayayala/cost-of-living-country-development.git